Donna Maione
Postdoctoral fellow (PhD)
Linnaeus University
donna.maione@lnu.se

Hanna Hofverberg
Associate professor (PhD)
Malmö University
hanna.hofverberg@mau.se

The craft and design of future cloth—an opening and exploration in recrafting for regenerative design

ABSTRACT

Roughly 100 billion garments are produced annually, and 60% becomes waste within the first year. With nearly 26 kilos of textiles being consumed annually by European, 11 kilos are being discarded. In response to this disposable behavior, a need to shift away from a mindset of consumption toward post-growth fashion habits are needed. This paper aims to build regenerative mindsets necessary for the challenges and possibilities of regenerative practices through design and craft education with the introduction of workshop methods. Two workshops were conducted in university settings, with first-year master's students in design and craft teacher students. The workshops revealed stories connecting past, present and future, with some focusing on traditional technique while others imagined the future of recrafted textiles. They also highlighted the need for continued methods in regenerative design and craft education, bridging traditional methods with future visions of textile recrafting.

Keywords: clothes; mending; regenerative practices; unlearning; teaching

INTRODUCTION

Clothing is a daily essential; we all wear clothes. But all of us also produce a lot of clothing waste, which has caused significant environmental problems (Brooks et al., 2017; Fletcher, 2016; Clean Clothes, 2018). For example, Clean Clothes (2018) reports that 100 billion garments are produced each year, with 60% becoming waste within the first year. The European Environmental Agency (EEA) (2020) indicates that Europeans consume nearly 26 kilos of textiles annually and discard about 11 kilos. Alarmingly, 87% of this discarded clothing is either incinerated or landfilled, according to the EEA. This unsustainable pattern of clothing consumerism and textile waste urgently needs to be addressed.

There are several initiatives minimizing clothing waste by changing systems, collectively and individually. For example, in Sweden, a new law prohibits discarding clothes as burnable waste (SFS, 2020): Instead, clothing waste must be sorted at the source (Assarsson, 2025). However, recycling centers are at a loss with what to do with the high volume of collected clothes (SVT, 2025). Another example of an initiative

to manage clothing waste is by collecting whole and clean clothes and sending them to those in need in other countries. However, these clothes are often inappropriate for the end user and end up in landfills or beaches in Ghana (Wennman & Lindberg, 2023). Both examples propose a solution that maintains the notion of unlimited materials and resources in what can be described as a cycle of taking, making, and wasting (Maione, 2023), that describes the linear design process that shifts the burden of waste. The waste is not minimized; it is just transferred to another place, out of sight of consumers.

To stop this unsustainable pattern of clothing waste, design research suggests shifting away from individualistic consumption toward post-growth fashion. This approach emphasizes the personal significance people find in their clothing (Fletcher, 2016) and encourages viewing clothes as more than mere "objects" for purchase and wear. Instead, clothes should be seen as integral parts of people's lives that transcend consumerism (Fry and Nocek, 2021).

While the consumers are partly responsible for the longevity of garments (Mead, 2015; Sekules, 2020), makers and designers must implement recrafting and repairing in their practices. To train students in recrafting and repairing as regenerative skills, alongside traditional design skills, can help us address the unsustainable waste stream of clothing (Ellen MacArthur Foundation, 2017; Maione, 2023). Regenerative design must thus become a common focus of any craft and design education.

In this paper, we argue for a shift away from the linear process and explore new ways of imagining plural futures for clothing in the context of design and craft education. In addition to focusing on the teaching and learning of technical repair methods, we are particularly interested in the stories of mending and the evolution of the material:

"By designing engagements with clothing repair and mending, shifting the focus to a more sustainable and post-capitalist approach can be achieved through intergenerational design practices that recognize the need for a hybrid material flow and multiple scales of production, ranging from local circular models to everyday mending practices." (Maione, 2023, 205)

We pose the question of how educational workshops can help to build regenerative mindsets in the context of clothing. As a starting point for our explorations, we report on two regenerative design and repair workshops as case studies on regenerative design through handcraft.

THEORY AND METHOD

Theoretically, the paper draws on the concept of regenerative design as a counternarrative to hyper-consumerism, particularly with clothing. The principle of regenerative design goes beyond the concept of "do no harm" or the sustainability of Earth's resources to include reparative generation as a net positive (Mang & Reed, 2012) to fix the harms created by industrialization.

To acknowledge regenerative design as a needed practice in teaching, we turn to stories of mending, which draw on Tim Ingold's notion of stories emerging from specific practices (Ingold 2013, cf. Hofverberg 2020). Ingold uses a tool as an example, suggesting that a tool must be imbued with a story that the practitioner needs to know and understand to recognize and use it properly. From Ingold's perspective, storytelling is not just a verbal act but also encompasses actions, implying that knowing how to use a tool involves engaging with the world in a meaningful way. Further, according to Ingold, tools do not come into the world with their meanings fully formed, nor are they invented on the spot without considering past practices. Instead, the stories emerge as a co-creating process recognized through the circumstances of the present in conjunction with the past within this specific context. In this paper, we explore clothing stories, emphasizing the practice of clothing and stories in use and reuse

and remaking. We recognize the conjunctions with past experiences of clothing—that we all have an experience of mass consumption, leading to wasted materials, which we need to unlearn when teaching stories of mending through craft. Furthermore, based on Ingold's suggestion that the meaning of tools emerges in the experience of making, we hypothesize that the repaired textile objects through handcraft bring the emergence of new meanings that are seeds toward new mindsets and ways of designing and crafting, contributing to regenerative practices and knowledge.

We review two workshops as case studies conducted in separate university settings. The first involved first-year master's students in design, while the second engaged craft teacher students. Both groups were tasked with bringing two items: a garment or textile in need of hand care or mending and a curious mind ready for exploration.

The two-hour workshop commenced with a brief introduction, underscoring the necessity of regenerative practices for designers, especially those navigating complex systems and engaging with others. The key idea was that reparative actions can bring about significant change, while also recognizing what is possible and what isn't in both reparative and regenerative design in our work, whether it's physical or not.

Following the introduction, a brief meditation session was conducted. We asked the participants to broaden their awareness using various senses. With their textile piece in front of them, they were asked to engage in a reflective process: to sit with their item in silence, look at the material's hand-feel and texture, think about its past, and examine traces from its history. "What stories emerge from traces of wear? We ended up asking participants to imagine the potential futures for the items in front of them. "What is the material telling you?"

RESULTS

A common outcome of the workshops revolved around stories connected to the past, such as tracing memories of items. A greater divergence of stories was more evident in the responses about the future. The design students told stories with a clearer vision of the future cloth, imagining their recrafted items, fixing tears (FIGURE 1), altering garments (FIGURE 2), making them desirable to wear again. Conversely, the craft students focused on a particular technique they would use on the garment (FIGURE 3). Here, the curiosity with the unknown diverged in the two workshops. This reminded us of the divergent mindsets of the designer and crafts person, where the unknown may be the place where design creates something that is not already in existence, while craft cherishes skills that focus on traditions.





FIGURES 1 AND 2. Left: Visible mending using hand crochet on damaged sweater at the cuff and sleeve area. A contrast color was chosen to add a design detail and to include the student's favorite color. Right: A found garment was restyled to be more in line with the student's fashion sense.





FIGURES 3 AND 4: Left: Participant focuses on craft techniques to mend the garments. Right: Participant has used blanket stitches to mend the pocket and explores the wear and tear of a pant leg before envisioning a future solution.

Beyond the stories connected to the past and future, stories emerged in conversation in the presence of handcrafting. Some participants crafted in silence; others were in conversation. One of the craft students expressed the active presence while mending as a "silence in conversation," highlighting that the participants continued their conversation even though they were not speaking. The design students, however, were more talkative, initiating conversations to learn about each other, as this was at the beginning of a new school year with an international student base. In both workshops, the slowness of hand stitching allowed space for contemplation and reflection between participants, material, and process, as well as a distraction from the newness of each other.

DISCUSSION

Pedagogical approaches centered on recrafting highlight the need for more time-intensive, slow handwork methods in design and craft education. To reshape the roles of designers and crafters, their education must help develop reparative craft skills alongside design abilities, bridging traditions with

future visions of textile recrafting. The shift from creating new objects to focusing on repair as a design method requires design education changes to include new tracks that lead to professional roles serving the emergent repair industry.

In particular, the results show that workshops for regenerative design must not only be future-oriented but also rooted in the past and present. In the workshops there are two stories emerging: a craft- story, which focuses on tradition and is a present-oriented activity, and the mending-story which bridges memories of items and future visions.

Further discussion on learning and unlearning is needed to bring into the future education and training for both craft and design. For example, while the skills for recrafting may resemble traditional crafts using needles and thread or yarns and needles, this shift necessitates embracing diverse paths into the future. In design education, how might design embrace making and craft as a method for creating the future of things, products, and services to regenerate the waste created?

REFERENCES

Assarsson, S. (2025, January 2). Sverige Radio [Broadcast]. https://shorturl.at/7bHvW

Brooks, A., Fletcher, K., Francis, R. A., Dulcie Rigby, E. & Roberts, T. (2017). Fashion, Sustainability, and the Anthropocene. *Utopian Studies*, 28,(3), 428-504. DOI: 10.5325/utopianstudies.28.3.0482

Clean Clothes (2018). Fashions Problem. Waste and pollution. https://cleanclothes.org/fashions-problems/waste-and-pollution

Ellen MacArthur Foundation. (2017). *A new textiles economy: Redesigning fashion's future*. http://www.ellenmacarthurfoundation.org/publications

European Environmental Agency (2020). The impact of textile production and waste on the environment. https://www.europarl.europa.eu/news/en/headlines/society/20201208STO93327/the-impact-of-textileproduction-and-waste-on-the-environment-infographics

Fletcher, K. (2016). *Craft of use: Post-growth fashion*. Routledge

Fry, T., & Nocek, A. (Eds.). (2021). *Design in crisis: New worlds, philosophies and practices*. Routledge, Taylor & Francis Group.https://doi.org/10.4324/9781003021469

Hofverberg, H. (2020). 'Crafting sustainability. Student's encounter with materiality'. Environmental Education Research, 26(9), 1281-1293.

Ingold, T. (2013). Making— *Anthropology, archaeology, art and architecture*. London: Routledge Taylor and Francis Group

Maione, D. (2023). *Recrafting futures: Post-material transformations toward clothing longevity.* Diss. Carnegie Mellon University.

Mang, P., & Reed, B. (2012). Designing from place: A regenerative framework and methodology. *Building Research & Information*, *40*(1), 23–38. https://doi.org/10.1080/09613218.2012.621341

Mead, D. (2015). *Design for Repair; Things can be fixed*. Cooper Hewitt, Smithsonian Design Museum. https://www.cooperhewitt.org/publications/design-for-repair/

Sekules, K. (2020). *Mend! A refashioning manual and manifesto*. Penguin Books.

SVT (2025). March 11. *Efter nya lagen: Värdelösa strumpor skickas till Litauen.* https://www.svt.se/nyheter/lokalt/skane/efter-nya-lagen-vardelosa-strumpor-skickas-till-litauen

Svensk författningssamling (SFS). (2020). Avfallsförordning (2020:614). [Retrieved from: https://www.riksdagen.se/sv/dokument-och-lagar/dokument/svensk-forfattningssamling/avfallsforordning-2020614_sfs-2020-614/].

Wennman, M. & Lindberg, L. (2023) June 11. Här dumpas kläderna du återvinner. *Aftonbladet.* https://www.aftonbladet.se/nyheter/a/O8PAyb/har-dumpas-h-m-kladerna-du-atervinner